

JOINS AND SET OPERATIONS

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SEC : 7

```
1 • CREATE DATABASE joins;
2 • USE joins;
3 ❌ Mohith Sai Chandra (2520030028)
4
5 -- CREATE TABLES
6 DROP TABLE IF EXISTS class_info;
7 • DROP TABLE IF EXISTS class;
8
9 • CREATE TABLE class (
10     id INT,
11     name VARCHAR(30)
12 );
13
14 • CREATE TABLE class_info (
15     id INT,
16     address VARCHAR(30)
17 );
18
```

Output

❏ Action Output

#	Time	Action	Message
✓ 1	20:58:24	CREATE DATABASE joins	1 row(s) affected
✓ 2	20:58:24	USE joins	0 row(s) affected
⚠ 3	20:58:28	DROP TABLE IF EXISTS class_info	0 row(s) affected, 1 warning(s)
⚠ 4	20:58:28	DROP TABLE IF EXISTS class	0 row(s) affected, 1 warning(s)
✓ 5	20:58:32	CREATE TABLE class (id INT, name VARCHAR(30))	0 row(s) affected
✓ 6	20:58:35	CREATE TABLE class_info (id INT, address VARCHAR(30))	0 row(s) affected

```

19  -- INSERT DATA
20  •  INSERT INTO class VALUES
21      (1,'abhi'),(2,'adam'),(3,'alex'),(4,'anu'),(5,'ashish');
22
23  •  INSERT INTO class_info VALUES
24      (1,'DELHI'),(2,'MUMBAI'),(3,'CHENNAI'),
25      (7,'NOIDA'),(8,'PANIPAT');
26
27  -- 1. CROSS JOIN
28  •  SELECT * FROM class
29      CROSS JOIN class_info;
30

```

Result Grid   Filter Rows: Export:  Wrap Cell Content: 

	id	name	id	address
▶	5	ashish	1	DELHI
	4	anu	1	DELHI
	3	alex	1	DELHI
	2	adam	1	DELHI
	1	abhi	1	DELHI
	5	ashish	2	MUMBAI
	4	anu	2	MUMBAI

Result 1 x

Output

 Action Output ▼

	#	Time	Action	Message
✓	1	20:59:03	INSERT INTO class VALUES (1,'abhi'),(2,'adam'),(3,'alex'),(4,'anu'),(5,'ashish')	5 row(s) affected Records: 5
✓	2	20:59:03	INSERT INTO class_info VALUES (1,'DELHI'),(2,'MUMBAI'),(3,'CHENNAI'), (7,'NOID...	5 row(s) affected Records: 5
✓	3	20:59:17	SELECT * FROM class CROSS JOIN class_info LIMIT 0, 1000	25 row(s) returned

```

31  -- 2. INNER JOIN
32  •  SELECT c.name, ci.address
33      FROM class c
34      INNER JOIN class_info ci
35      ON c.id = ci.id;
36

```

Result Grid			Filter Rows:	Export:	Wrap Cell Content:
	name	address			
▶	abhi	DELHI			
	adam	MUMBAI			
	alex	CHENNAI			

Result 2 ×

Output

Action Output

#	Time	Action	Message
✓ 1	21:00:07	SELECT c.name, ci.address FROM class c INNER JOIN class_info ci ON c.id = ci.id ...	3 row(s) returned

```

37  -- 3. NATURAL JOIN
38  •  SELECT * FROM class
39      NATURAL JOIN class_info;
40

```

Result Grid				Filter Rows:	Export:	Wrap Cell Content:
	id	name	address			
▶	1	abhi	DELHI			
	2	adam	MUMBAI			
	3	alex	CHENNAI			

Result 3 ×

Output

Action Output

#	Time	Action	Message
✓ 1	21:00:34	SELECT * FROM class NATURAL JOIN class_info LIMIT 0, 1000	3 row(s) returned

```

41      -- 4. LEFT JOIN
42      • SELECT *
43      FROM class c
44      LEFT JOIN class_info ci
45      ON c.id = ci.id;
46

```

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	id	name	id	address
▶	1	abhi	1	DELHI
	2	adam	2	MUMBAI
	3	alex	3	CHENNAI
	4	anu	NULL	NULL
	5	ashish	NULL	NULL

Result 4 ×				
Output				
Action Output				
#	Time	Action	Message	
✓ 1	21:01:11	SELECT * FROM class c LEFT JOIN class_info ci ON c.id = ci.id LIMIT 0, 1000	5 row(s) returned	

```

47      -- 5. LEFT JOIN + NULL
48      • SELECT *
49      FROM class c
50      LEFT JOIN class_info ci
51      ON c.id = ci.id
52      WHERE ci.id IS NULL;
53

```

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	id	name	id	address
▶	4	anu	NULL	NULL
	5	ashish	NULL	NULL

Result 5 ×				
Output				
Action Output				
#	Time	Action	Message	
✓ 1	21:01:30	SELECT * FROM class c LEFT JOIN class_info ci ON c.id = ci.id WHERE ci.id IS N...	2 row(s) returned	

```

54      -- 6. RIGHT JOIN
55      • SELECT *
56      FROM class c
57      RIGHT JOIN class_info ci
58      ON c.id = ci.id;
59

```

Result Grid				
	id	name	id	address
▶	1	abhi	1	DELHI
	2	adam	2	MUMBAI
	3	alex	3	CHENNAI
	NULL	NULL	7	NOIDA
	NULL	NULL	8	PANIPAT

Result 6 ×				
Output				
Action Output				
#	Time	Action	Message	
✓ 1	21:02:10	SELECT * FROM class c RIGHT JOIN class_info ci ON c.id = ci.id LIMIT 0, 1000	5 row(s) returned	

```

60      -- 7. RIGHT JOIN + NULL
61      • SELECT *
62      FROM class c
63      RIGHT JOIN class_info ci
64      ON c.id = ci.id
65      WHERE c.id IS NULL;

```

Result Grid				
	id	name	id	address
▶	NULL	NULL	7	NOIDA
	NULL	NULL	8	PANIPAT

Result 7 ×				
Output				
Action Output				
#	Time	Action	Message	
✓ 1	21:02:38	SELECT * FROM class c RIGHT JOIN class_info ci ON c.id = ci.id WHERE c.id IS N...	2 row(s) returned	

```

67      -- 8. FULL OUTER JOIN
68  •    SELECT *
69      FROM class c
70      LEFT JOIN class_info ci ON c.id = ci.id
71      UNION
72      SELECT *
73      FROM class c
74      RIGHT JOIN class_info ci ON c.id = ci.id;

```

Result Grid				
		Filter Rows:		Export:  Wrap Cell Content: 
	id	name	id	address
▶	1	abhi	1	DELHI
	2	adam	2	MUMBAI
	3	alex	3	CHENNAI
	4	anu	NULL	NULL
	5	ashish	NULL	NULL
	NULL	NULL	7	NOIDA
	NULL	NULL	8	PANIPAT

Result 8 ×			
Output			
Action Output			
#	Time	Action	Message
✓ 1	21:03:12	SELECT * FROM class c LEFT JOIN class_info ci ON c.id = ci.id UNION SELECT * ...	7 row(s) returned

```

76      -- 9. UNION
77      • SELECT id, name FROM class
78      UNION
79      SELECT id, address FROM class_info;

```

Result Grid		
	Filter Rows:	Export: Wrap Cell Content:
	id	name
▶	1	abhi
	2	adam
	3	alex
	4	anu
	5	ashish
	1	DELHI
	2	MUMBAI
	3	CHENNAI
	7	NOIDA
	8	PANIPAT

Result 9 ×

Output

Action Output

#	Time	Action	Message
✓ 1	21:03:37	SELECT id, name FROM class UNION SELECT id, address FROM class_info	10 row(s) returned

```

81      -- 10. UNION ALL
82      • SELECT id, name FROM class
83      UNION ALL
84      SELECT id, address FROM class_info;
85

```

Result Grid		
	Filter Rows:	Export: Wrap Cell Content:
	id	name
▶	1	abhi
	2	adam
	3	alex
	4	anu
	5	ashish
	1	DELHI
	2	MUMBAI
	3	CHENNAI
	7	NOIDA
	8	PANIPAT

Result 10 ×

Output

Action Output

#	Time	Action	Message
✓ 1	21:04:12	SELECT id, name FROM class UNION ALL SELECT id, address FROM class_info	10 row(s) returned

```
86      -- 11. INTERSECT
87      • SELECT id, name FROM class
88      INTERSECT
89      ✖ SELECT id, address FROM class_info;
90
```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
id	name			

Result 11 ✖			
Output			
Action Output			
#	Time	Action	Message
✓ 1	21:04:46	SELECT id, name FROM class INTERSECT SELECT id, address FROM class_info	0 row(s) returned


```

91  -- 12. MINUS
92  •  SELECT c.id, c.name
93      FROM class c
94  WHERE NOT EXISTS (
95      SELECT 1
96      FROM class_info ci
97      WHERE ci.id = c.id
98  );
99

```

Result Grid			Filter Rows:	Export:	Wrap Cell Content:
	id	name			
▶	4	anu			
	5	ashish			

class 12 ×			
Output			
Action Output			
#	Time	Action	Message
✓ 1	21:05:13	SELECT c.id, c.name FROM class c WHERE NOT EXISTS (SELECT 1 FROM...	2 row(s) returned

```

100      -- 13. CASE
101 •    SELECT c.id, c.name,
102      CASE
103          WHEN ci.address IS NULL THEN 'Address Missing'
104          ELSE 'Address Available'
105      END AS Status
106      FROM class c
107      LEFT JOIN class_info ci
108      ON c.id = ci.id;

```

Result Grid			
Filter Rows:			
Export:  Wrap Cell Content: 			
	id	name	Status
▶	1	abhi	Address Available
	2	adam	Address Available
	3	alex	Address Available
	4	anu	Address Missing
	5	ashish	Address Missing

Result 13 x			
Output			
Action Output			
#	Time	Action	Message
✓ 1	21:05:31	SELECT c.id, c.name, CASE WHEN ci.address IS NULL THEN 'Address Missing' ...	5 row(s) returned